When Reviewers Lock Horn: Finding Disagreement in Scientific Peer Reviews

Sandeep Kumar†, Tirthankar Ghosal‡, Asif Ekbal†
†Indian Institute of Technology Patna, India
‡National Center for Computational Sciences, Oak Ridge National Laboratory, USA
†(sandeep_2121cs29,asif)@iitp.ac.in
‡ghosalt@ornl.gov

Abstract

To this date, the efficacy of the scientific publishing enterprise fundamentally rests on the strength of the peer review process. The journal editor or the conference chair primarily relies on the expert reviewers' assessment, *identify points of agreement and disagreement* and try to reach a consensus to make a fair and informed decision on whether to accept or reject a paper. However, with the escalating number of submissions requiring review, especially in top-tier Artificial Intelligence (AI) conferences, the editor/chair, among many other works, invests a significant, sometimes stressful effort to mitigate reviewer disagreements. Here in this work, we introduce a novel task of automatically identifying contradictions among reviewers on a given article. To this end, we introduce *ContraSciView*, a comprehensive review-pair contradiction dataset on around 8.5k papers (with around 28k review pairs containing nearly 50k review pair comments) from the open review-based ICLR and NeurIPS conferences. We further propose a baseline model that detects contradictory statements from the review pairs. To the best of our knowledge, we make the first attempt to identify disagreements among peer reviewers automatically. We make our dataset and code public for further investigations[1].

1 Introduction

Despite being the widely accepted standard for validating scholarly research, the peer-review process has faced substantial criticism. Its perceived lack of transparency (Wicherts, 2016; Parker et al., 2018), sometimes being biased (Stelmakh et al., 2021, 2019), arbitrariness (Brezis and Birukou, 2020), inconsistency (Shah et al., 2018; Langford and Guzdial, 2015), being regarded as a poorly defined task (Rogers and Augenstein, 2020a) and failure to recognize influential papers (Freyne et al., 2010) have

[1]https://github.com/sandeep82945/Contradiction-in-Peer-Review and https://www.iitp.ac.in/~ai-nlp-ml/resources.html#ContraSciView

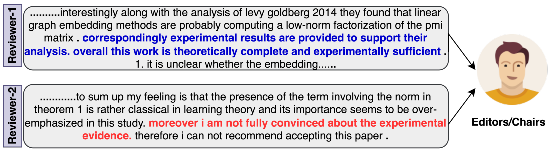

Figure 1: An example of contradiction among reviewers.

all been subjects of concern within the scholarly community. The rapid increase in research article submissions across different venues has caused the peer review system to undergo a huge amount of stress (Kelly et al., 2014; Gropp et al., 2017). The role of an editor in scholarly publishing is crucial (Hames, 2012). Typically, editors or chairs have to manage a multitude of responsibilities. These include finding expert reviewers, assigning review tasks, mediating disagreements among reviewers, ensuring reviews are received on time, stepping in when reviewers are not responsive, making informed decisions, and maintaining communication with authors, among other duties. Simultaneously, they must ensure the validity and quality of the reviews to make an informed decision. However, the complexity of scholarly discourse and inherent subjectivity within research interpretation sometimes leads to conflicting views among peer reviewers (Bornmann and Daniel, 2009; Borcherds and Editor, 2017). For instance, in Figure 1, Reviewer 1 regards the evidence as both strong and sufficient, reinforcing the paper's theoretical soundness. Conversely, Reviewer 2 remains skeptical of this evidence, underscoring their differing perspectives on the soundness of the paper.

Feedback between authors and reviewers can help improve the peer review system (Rogers and Augenstein, 2020b). However, reviewer disagreement can create confusion for authors and editors as they try to address reviewer feedback. There are various guidelines or suggestions that editors consider in order to resolve the conflicting reviews (Borcherds and Editor, 2017). Given this volume, it is challenging for editors to identify contradictions manually. Our current investigation can assist in pinpointing these discrepancies and help editors make informed decisions.

It is to be noted that this AI-based system aims to aid editors in identifying potential contradictions in reviewer comments. While it provides valuable insights, it is not infallible. Editors should use it as a supplementary tool, understanding that not all contradictions may be captured and manual review remains crucial. They should make decision with careful analysis beyond the system's recommendations.

Our contributions are *three-fold*: 1) We introduce a novel task: identifying contradictions/disagreement within peer reviews. 2) To address the task, we create a novel labeled dataset of around 8.5k papers and 25k reviews. 3) We establish a baseline method as a reference point for further research on this topic.

2 Related Work

Artificial Intelligence (AI) has been applied in recent years to the realm of scientific peer review with the goal of enhancing its efficacy and quality (Checco et al., 2021; Ghosal et al., 2022). These applications span a diverse range of tasks, including decision prediction (Kumar et al., 2022; Ghosal et al., 2019), rating prediction (Li et al., 2020; Kang et al., 2018a), sentiment analysis (Kumar et al., 2021; Chakraborty et al., 2020; Kang et al., 2018b), argument mining (Hua et al., 2019; Cheng et al., 2020), and review summarization (Xiong, 2013). In this work, we aim to broaden the scope of AI's usefulness by assisting the editor in determining the disagreement between reviewers. Contradiction detection is not new. Alamri and Stevenson (2015) use linguistic features and support vector machines (SVMs) to detect contradictions in scientific claims, while Badache et al. (2018b) employ review polarity to predict contradiction intensity in reviews. Li et al. (2018) combine sentiment analysis with contradiction detection for Amazon reviews. Meanwhile, Lendvai and Reichel (2016) employ textual similarity features to classify contradictions in rumors, and Li et al. (2017) use contradiction-specific word embeddings for sentence pairs. Tan et al. (2019) propose a dual attention-based gated recurrent unit for conflict detection. Recent advances in Natural Language Inference (NLI) have brought forth a category of textual entailment, categorizing text pairs into entailment, neutral, or contradiction (Chen et al., 2017; Gong et al., 2018). Well-known transformer-based models, such as BERT (Devlin et al., 2019) and its variants leading to state-of-the-art performance.

Venues	Papers	#Reviews	# Pairs
ICLR	5,096	14,711	14,383
NeurIPS	3,486	11,114	14,114
Total	8,582	25,825	28,497

Table 1: Dataset statistics

As far as we know, contradiction detection in peer review has never been studied. Contradiction detection in peer reviews is complex and requires domain knowledge of the subject. It is not straightforward to detect contradictions in peer reviews because reviewers often have different writing styles and approaches to commenting. We believe that our work can significantly contribute to the peer review process.

3 Dataset

3.1 Dataset Collection

We utilize a subset of 8,582 out of 8,877 papers from the extensive ASAP-Review dataset (Yuan et al., 2021). The dataset comprises reviews from the ICLR (2017-2020) and NeurIPS (2016-2019) conferences. Each review is labeled with eight aspect categories: *(Motivation, Clarity, Soundness, Substance, Originality, Meaningful Comparison, Replicability, and Summary)* along with their sentiment *(Positive/Negative)* except Summary.

3.2 Dataset Pre-processing

We define *review* is as a collection of comments/sentences written by one reviewer. Formally, we can represent it as a list:

$$R = \{\mathrm{cmt}_1, \mathrm{cmt}_2, \ldots\}$$

A *review pair* takes two such lists, one from Reviewer1 and one from Reviewer2. It can be represented as:

$$RP = \{R_1, R_2\}$$

Lastly, a *review pair comment* selects one comment from each reviewer and forms a pair. It is a set of such pairs:

$$RPC = \{(\text{cmt from } R_1, \text{cmt from } R_2), \ldots\}$$

To make it easier to annotate, we first create pairs of reviews of papers. Suppose, if there are n number of reviews in a paper then we create $\binom{n}{2}$ pairs resulting in a total of around 28k pairs. Detailed statistics regarding this dataset can be found in Table 1. We follow the contradiction definition of Badache et al. (2018a). According to this definition, a contradiction exists between two review pairs when any review pair comments, denoted as ra_1 and ra_2, contain a common aspect category but convey opposite sentiments. Therefore, we categorize those review pairs as *no contradiction* if none of their comments share the same aspect with opposing sentiments. This labeling process resulted in 17,095 pairs of reviews. For the remaining review pairs, we compile a list of review pair comments that share the same aspect but express opposing sentiments, and we designate these for human annotation. *Finally, we annotate a total of 50,303 pair of review pair comments.*

4 Annotating for Contradiction

4.1 Annotation Guidelines

We follow the contradiction definition by De Marneffe et al. (2008) for our annotation process where they define contradiction as: *"ask yourself the following question: If I were shown two contemporaneous documents, one containing each of these passages, would I regard it as very unlikely that both passages could be true at the same time? If so, the two contradict each other"*. We offer multiple examples from different aspect categories that may contain conflicting reviews to assist and direct the annotators. Figure 4 illustrates the flowchart that represents our annotation procedure. For the detailed annotation guidelines, please refer to Appendix A.

4.2 Annotator Training

Given the complexity of the reviews and their frequent use of technical terminology, we had six doctoral students, each with four years of experience in scientific research publishing. To facilitate their training, two experts with more than ten years of experience in scientific publishing annotated 1,500 review pairs from a selection of random papers, following our guidelines. Our experts convened to discuss and reconcile any discrepancies in their annotations. The initial dataset comprises 227 pairs with contradictions and 1273 pairs without contradiction comments. We randomly selected 100 review pairs from this more extensive set to train our annotators, ensuring both classes are equally represented. Upon completion of this round of annotation, we reviewed and corrected any misinterpretations with the annotators, further refining their training and enhancing the clarity of the annotation guidelines. To evaluate the effectiveness of the initial training round, we compiled another 80 review pairs from both classes drawn from the remaining review pairs. From the second round on wards, most annotators demonstrated increased proficiency, accurately annotating at least 70% of the contradictory cases.

4.3 Annotation Process

We regularly monitored the annotated data, placing emphasis on identifying and rectifying inconsistencies and cases of confusion. We also implemented an iterative feedback system that continuously aimed to refine and improve the annotation process. In cases of conflict or confusion, we consulted experts to make the final decision. Following the annotation phase, we obtained an average inter-annotator agreement score of 0.62 using Cohen's kappa (Cohen, 1960), indicating a substantial consensus among the annotators.

4.4 Annotator's Pay

We compensated each annotator based on standard salaries in India, calculated by the hours they worked. The appointment and salaries are governed by the standard practices of our university. We chose not to pay per review pair because the time needed to understand reviews varies due to their complexity, technical terms, and the annotator's familiarity with the topic. Some reviews are also extensive, requiring more time to comprehend. Hence, basing pay on review pairs could have compromised annotation quality. To ensure accuracy and prevent fatigue, we set a daily limit of 6 hours for annotators.

4.5 Final Dataset

Figure 2 displays the annotation statistics. We observed that the majority of disagreements among reviewers pertain to the paper's clarity. Such disagreements could stem from differences in reviewers' expertise, domain knowledge (a reviewer unfamiliar with the domain might find it hard to grasp the content), language proficiency (some reviewers might struggle with English, while others are fluent), or interest level (a disinterested reader might

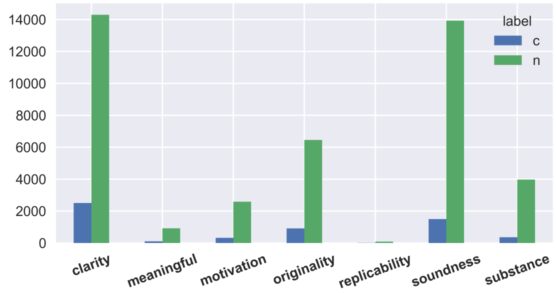

Figure 2: Comparison of Contradiction by Aspect; y axis: count of review comment, x: aspect category, c: contradiction, n: non-contradiction

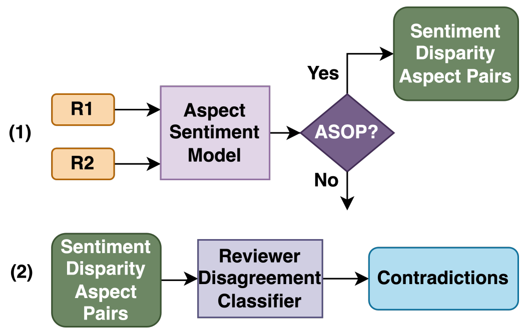

Figure 3: Flowchart of our proposed baseline. Here, R1 and R2 represent the two reviews. The baseline consists of two steps: (1) extracting the SDAPs and (2) determining whether these SDAPs are contradictions ; ASOP stands for "Is Aspect same and sentiment opposite?"

find the paper difficult to engage with). Discrepancies regarding replicability and meaningful comparison are notably fewer, likely because these topics are less frequently commented on.

5 Baseline System Description

We describe the flow of our proposed baseline setup through the flowchart in Figure 3. The initial input involves a pair of reviews for a given paper, which are subjected to the Aspect Sentiment Model. This model classifies the aspect and sentiment of each review comment/sentence within the reviews. Subsequently, we identify pairs of comments that, while sharing the same aspects, exhibit differing sentiments; we term it as Sentiment Disparity Aspect Pair (SDAP). As a final step, the SDAPs are then passed to the Contradiction Classifier in order to classify whether these pairs of review comments are contradictory or not. We describe the Aspect Sentiment Model and Contradiction Detection Model in details as follows:

Aspect Sentiment Model: Aspect and sentiment in peer review have been studied as a multi-task model (Kumar et al., 2021). A review pair comment can have different sentiments corresponding to multiple aspect categories. So, we utilize Multi-Instance Multi-Label Learning Network (MIMLLN) (Li et al., 2020) which uniquely identifies sentences as bags and words as instances. It functions as a multi-task model, which performs both Aspect Category Detection (ACD) and Aspect Category Sentiment Analysis (ACSA). Given a sentence and its aspect categories, it predicts sentiment by identifying key instances, and aggregating these to obtain the sentence's sentiment towards each category. We discuss MIMLLN in more detail in Appendix B. We trained MIMLLN on the human-annotated ASAP-Review dataset for our task.

Reviewer Disagreement Classifier We use techniques from Natural Language Inference (NLI) sentence pair classification to identify reviewer disagreement, particularly contradictions from Sentence Dependency Pairs (SDPs). Unlike traditional NLI tasks that provide a three-category output of "entailment", "contradiction", and "neutral", we have adjusted the model to a two-category output system: "contradiction" and "non-contradiction". The latter category combines "entailment" and "neutral" labels, as our primary focus is on contradiction detection.

6 Results and Analysis

We discuss the implementation details in Appendix C. Table 2 reports the performance of the models when trained on our dataset. The results are of the Reviewer Disagreement Classifier, which is the final step in our proposed approach. We report macro average P, R, and F1 scores as the rarity of the contradiction class is of particular interest for this task (Gowda et al., 2021). RoBERTa large outperforms XLNet large by 0.7 points, RoBERTa base by 4.1 points, XLNet by 4.5 points, and DistilBERT by 7.5 points with respect to F1 score.

In order to analyze how models perform when trained on natural language inference datasets, we trained the models on the ANLI+ALL dataset and evaluated them on our test set. It was found that the models trained by combining datasets, i.e., SNLI, MNLI, FEVER, and ANLI A1+A2+A3, perform the best (Nie et al., 2020). We obtain the highest F1 score of 71.14 F1 with RoBERTa Large. The results show that all the models trained on our dataset outperform those trained on the existing datasets. This large increase in performance can